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Editors

Leonardo Valerio Noto

Francesco Alongi

Dario Treppiedi

Harris Vangelis

Dimitris Tigkas



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EWRA Editorial Office:

Iroon Polytechniou 9, 157 80, Athens, Greece

e-mail: puboffice@ewra.net

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Liquid and solid phase effects of elevated bioreactor dissolved oxygen

G. Posadas¹, K. Davis², S. Miller^{1*}

¹ Department of Civil Engineering, Boise State University, Boise, Idaho, U.S.A.

² EnviroTech Catalyst, LLC, Boise, ID 83702, U.S.A.

* e-mail: sondramiller1@boisestate.edu

Introduction

The concentration of dissolved oxygen (DO) in biological wastewater treatment systems plays a crucial role in both liquid- and solid-phase treatment dynamics, influencing microbial activity, nutrient removal, and biosolids production. This study examines the effects of elevated DO concentrations in an oxidation ditch induced by a high concentration DO infusion system retrofitted into a wastewater treatment plant (WWTP) in Greenleaf, Idaho, evaluating its impacts on both nutrient reduction and biosolids generation. Multi-correlation analyses indicate that elevated DO concentrations enhance nitrogen and phosphorus removal and contribute to increased sludge volume index (SVI), suggesting shifts in microbial community structure and sludge settling quality. Machine learning models reveal a statistically significant, albeit weak, negative correlation between elevated DO and biosolids generation. These findings underscore the dual impacts of increased DO, improving liquid-phase treatment efficiency while altering solid-phase biosolids generation, offering insight into optimizing bioreactor operation for enhanced sustainability.

Materials and methods

This study was conducted at the Greenleaf WWTP in Idaho, a 0.24 MGD facility retrofitted in January 2024 with a high concentration DO infusion system. The system pre-dissolves oxygen into return activated sludge (RAS) before reintroduction into the oxidation ditch.

Wastewater samples were collected over a one-year period (January 2024 to December 2024) from the raw influent, oxidation ditch effluent, RAS before and after DO infusion, and final effluent. The following parameters were monitored: DO, 5-day biochemical oxygen demand (BOD₅), chemical oxygen demand (COD), total nitrogen, total phosphorus, total suspended solids (TSS), volatile solids, sludge volume index (SVI), and adenosine triphosphate (ATP) fractions. On-site measurements for DO, pH, temperature, and ATP, were conducted using a Hach HQ4200 multi-meter and a LuminUltra QG21W test kit. Laboratory analyses followed Standard Methods for the Examination of Water and Wastewater (APHA 2017) and Hach Test 'N Tube (TNT) reagent kit procedures. Biosolids generation data was collected through conveyor runtimes and time-standardized weight monitoring, which was limited by reliance on intermittent operator records and variability in dewatering operations, leading to inconsistencies in measurement availability.

A multi-correlation analysis was performed to assess the impact of elevated DO concentrations on liquid-phase treatment processes using combined Pearson, Spearman, and Kendall coefficients. These analyses evaluated relationships between DO concentrations at the RAS return point and nutrient removal efficiencies, including changes in total nitrogen and total phosphorus. Sludge settleability trends were also analyzed, with a focus on the sludge volume index (SVI) and cellular ATP (cATP) as indicators of microbial activity.

A machine learning-based predictive modeling approach was used to interpolate the limited biosolids generation data. A Random Forest regression model was applied to identify key predictors influencing biosolids generation, while multiple imputation methods addressed missing data in conveyor runtimes. A secondary XGBoost regression model was implemented to refine biosolids generation estimates. Feature selection techniques were used to prioritize the most influential parameters, allowing for a data-driven assessment of the relationship between elevated DO levels and biosolids reduction.

Results and concluding remarks

The high-concentration DO infusion system significantly impacted both liquid-phase treatment and biosolids generation at the Greenleaf WWTP. Correlation analyses showed that increasing DO concentrations at the RAS return point (Figure 1) enhanced nitrogen and phosphorus removal efficiencies. Nitrate concentrations in the oxidation ditch effluent decreased with the increasing DO concentrations in the RAS return point ($r = -0.47$) which subsequently led to a decrease of total nitrogen in the final effluent. Similarly, phosphorus concentrations in the final effluent declined following system implementation ($r = -0.59$). A notable operational trade-off was observed in the sludge volume index (SVI), which increased with rising DO levels ($r = 0.55$), potentially due to microbial shifts negatively affecting sludge settleability.

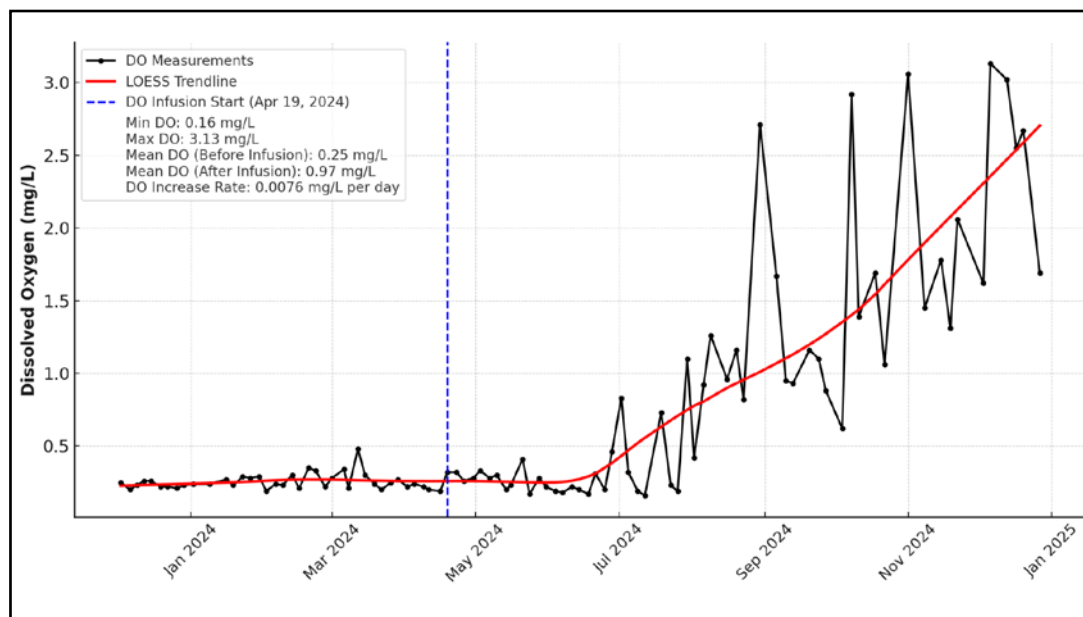


Figure 1. DO increase at RAS return point in oxidation ditch.

Machine learning imputed biosolids generation data revealed a weak statistically significant negative correlation ($r = -0.26$) between elevated DO and biosolids production. This suggests that higher DO concentration may contribute to reduced biosolids generation, likely by enhancing organic matter degradation before sludge separation. Feature selection methods further highlighted influent BOD ($r = 0.76$), RAS return point pH ($r = -0.72$), and oxidation ditch effluent 30-minute sludge volume ($r = -0.71$) as the strongest predictors of biosolids generation, underscoring the complexity of solids-phase responses to DO enrichment.

These findings demonstrate the dual-phase impacts of high-DO infusion in wastewater treatment. While enhanced oxygenation supports improved nutrient removal, it also introduces operational challenges such as increased SVI. Future research should focus on optimizing DO delivery strategies to balance nutrient removal efficiency with sludge settleability while integrating real-time monitoring systems for better control of liquid-phase and solid-phase treatment dynamics.

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